

Name: Muhammad Yahya

F/Name: Muhammad Hussain

ID: BAI-23F-030

Semester: 6th(B)

Program: BSAI(Morning)

Subject Lab: NLP

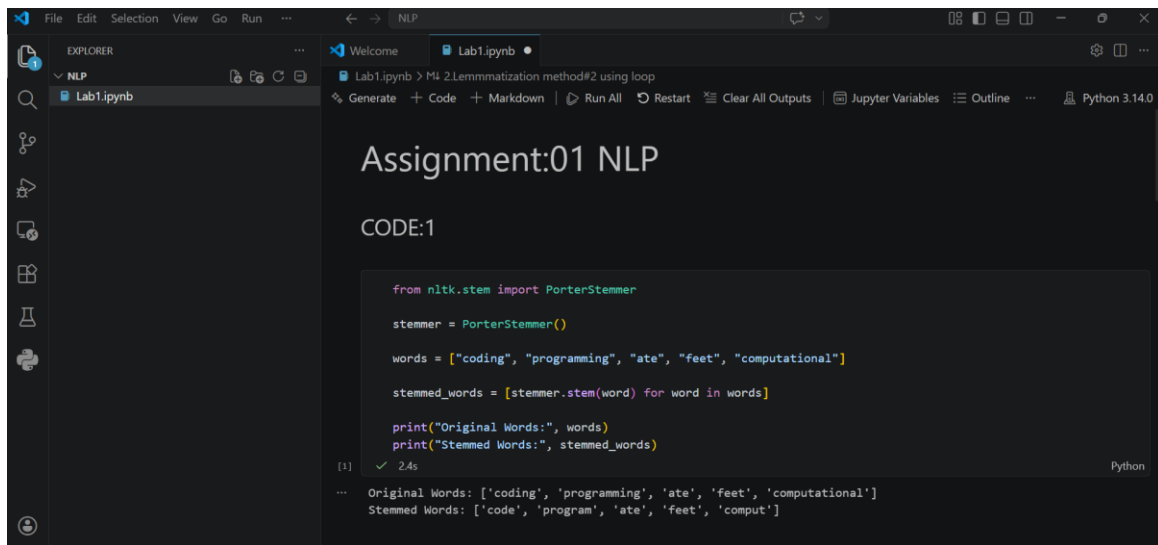
Teacher: Sir Usman

Assignment#01: NLP(LAB)

Requirements:

1. Change the variable "Words" and use your own words
2. Change the variable "Sentence" and use your own words

stemming



The screenshot shows a Jupyter Notebook titled "Assignment:01 NLP" in a VS Code editor. The notebook is running a Python script that demonstrates stemming using NLTK's PorterStemmer. The code defines a list of words, applies the stemmer to each word, and prints the original and stemmed words. The output shows the original words as a list of strings and the stemmed words as a list of strings with single quotes around them.

```
from nltk.stem import PorterStemmer

stemmer = PorterStemmer()

words = ["coding", "programming", "ate", "feet", "computational"]

stemmed_words = [stemmer.stem(word) for word in words]

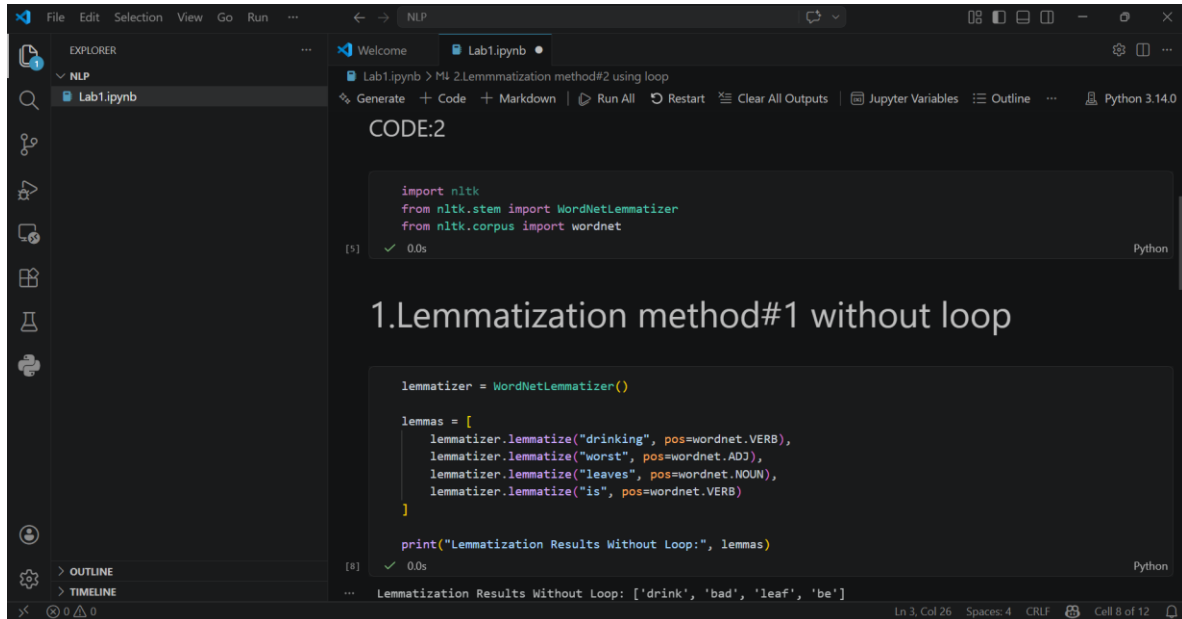
print("Original Words:", words)
print("Stemmed Words:", stemmed_words)
```

Output:

```
Original Words: ['coding', 'programming', 'ate', 'feet', 'computational']
Stemmed Words: ['code', 'program', 'ate', 'feet', 'comput']
```

Lemmatization

Method:1 Without Loop



The screenshot shows a Jupyter Notebook interface with a dark theme. The Explorer pane on the left shows a file named 'Lab1.ipynb'. The main editor area has a title bar 'Lab1.ipynb' and a subtitle 'M4 2.Lemmatization method#2 using loop'. The code cell is titled 'CODE:2' and contains the following Python code:

```
import nltk
from nltk.stem import WordNetLemmatizer
from nltk.corpus import wordnet
```

Below the code, the output of the cell is displayed:

```
1.Lemmatization method#1 without loop
```

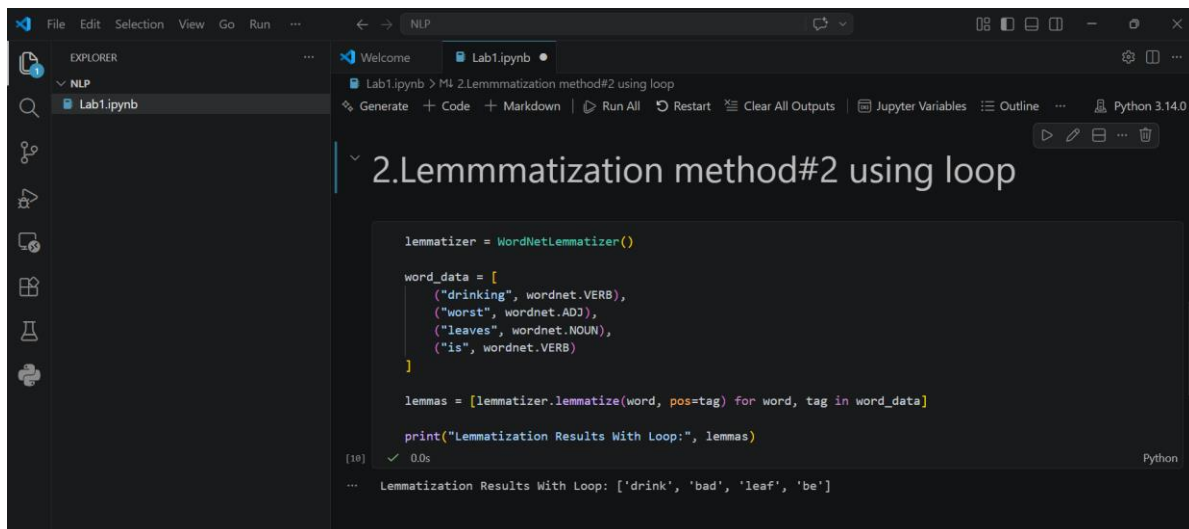
```
lemmatizer = WordNetLemmatizer()

lemmas = [
    lemmatizer.lemmatize("drinking", pos=wordnet.VERB),
    lemmatizer.lemmatize("worst", pos=wordnet.ADJ),
    lemmatizer.lemmatize("leaves", pos=wordnet.NOUN),
    lemmatizer.lemmatize("is", pos=wordnet.VERB)
]

print("Lemmatization Results Without Loop:", lemmas)
```

The output shows the lemmas: ['drink', 'bad', 'leaf', 'be'].

Method:2 Using Loop



The screenshot shows a Jupyter Notebook interface with a dark theme. The Explorer pane on the left shows a file named 'Lab1.ipynb'. The main editor area has a title bar 'Lab1.ipynb' and a subtitle 'M4 2.Lemmatization method#2 using loop'. The code cell is titled '2.Lemmatization method#2 using loop' and contains the following Python code:

```
lemmatizer = WordNetLemmatizer()

word_data = [
    ("drinking", wordnet.VERB),
    ("worst", wordnet.ADJ),
    ("leaves", wordnet.NOUN),
    ("is", wordnet.VERB)
]

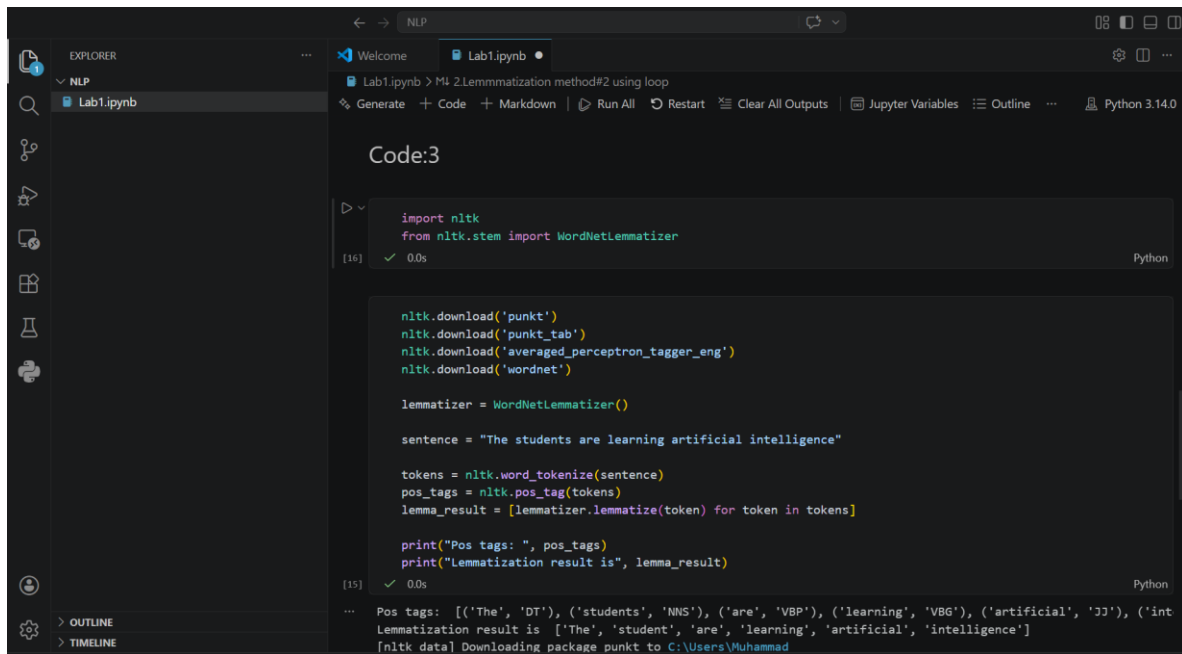
lemmas = [lemmatizer.lemmatize(word, pos=tag) for word, tag in word_data]

print("Lemmatization Results With Loop:", lemmas)
```

The output of the cell is displayed:

```
Lemmatization Results With Loop: ['drink', 'bad', 'leaf', 'be']
```

2. Change the variable "Sentence" and use your own words



The screenshot shows a Jupyter Notebook environment with a dark theme. The Explorer pane on the left shows a file named 'Lab1.ipynb'. The main editor area displays the following code:

```
Code:3

import nltk
from nltk.stem import WordNetLemmatizer

[16] ✓ 0.0s Python

nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('averaged_perceptron_tagger_eng')
nltk.download('wordnet')

lemmatizer = WordNetLemmatizer()

sentence = "The students are learning artificial intelligence"

tokens = nltk.word_tokenize(sentence)
pos_tags = nltk.pos_tag(tokens)
lemma_result = [lemmatizer.lemmatize(token) for token in tokens]

print("Pos tags: ", pos_tags)
print("Lemmatization result is", lemma_result)

[15] ✓ 0.0s Python
```

The output of the code is displayed below the cell:

```
Pos tags: [(['The', 'DT'), ('students', 'NNS'), ('are', 'VBP'), ('learning', 'VBG'), ('artificial', 'JJ'), ('int', 'int')]
Lemmatization result is ['The', 'student', 'are', 'learning', 'artificial', 'intelligence']
[nltk data] Downloading package punkt to C:\Users\Muhammad
```